

TaskForge - Laravel Learning Example Project

A team task management application built to demonstrate every major Laravel concept from the [learning curriculum](#).

Stack: Laravel 12 ? PHP 8.2+ ? SQLite/MySQL ? Sanctum ? Tailwind CSS ? Vite

Quick Start

```
# From taskforge directory
copy .env.example .env
php artisan key:generate

# SQLite (default - easiest)
# Ensure database/database.sqlite exists, or:
type nul > database\database.sqlite

php artisan migrate --seed
php artisan storage:link
php artisan serve
```

Visit **http://localhost:8000**

Account	Email	Password
Admin	admin@taskforge.test	password
Member	member@taskforge.test	password

Features & Laravel Concepts

Feature	Files to Study	Module
User auth (login/register)	AuthController, auth/* views	06
Dashboard with cached stats	DashboardService, DashboardController	13
Project CRUD	ProjectController, ProjectPolicy	03, 05
Task CRUD + filters	TaskController, Task model scopes	03, 05
Task assignments (M2M)	task_user migration, assignees()	05
Comments	Comment model, storeComment()	05
File attachments	Attachment, storeAttachment()	10
Form validation	StoreTaskRequest, etc.	09
Authorization policies	TaskPolicy, ProjectPolicy	06
Events & activity log	TaskCompleted, ActivityLog	12
Queued notifications	SendTaskDueReminder, TaskDueReminder	11, 12
REST API (Sanctum)	TaskApiController, TaskResource	07
Scheduled commands	SendDueReminders, routes/console.php	11
PHPUnit tests	tests/Feature/*	14
PHP Enums	app/Enums/*	01

Project Structure

```
app/
|-- Console/Commands/SendDueReminders.php
|-- Enums/                                # TaskStatus, TaskPriority, UserRole
|-- Events/TaskCompleted.php
|-- Http/
|   |-- Controllers/                     # Web controllers
|   |-- Controllers/Api/                 # API controllers
|   |-- Requests/                         # Form validation
|   |-- Resources/                       # API JSON transformers
|-- Jobs/SendTaskDueReminder.php
|-- Listeners/LogTaskCompletion.php
|-- Models/                             # Eloquent models
|-- Notifications/TaskDueReminder.php
|-- Policies/                           # Authorization
|-- Services/DashboardService.php
database/
|-- migrations/
|-- factories/
|-- seeders/DatabaseSeeder.php
resources/views/
|-- layouts/app.blade.php
|-- auth/
|-- projects/
|-- tasks/
|-- dashboard.blade.php
routes/
|-- web.php
|-- api.php
|-- console.php
tests/
|-- Feature/AuthTest.php
|-- Feature/TaskTest.php
|-- Feature/Api/TaskApiTest.php
deploy/
|-- nginx.conf
|-- supervisor.conf
|-- github-actions.yml
```

Common Commands

php artisan serve	# Dev server
php artisan test	# Run tests
php artisan queue:work	# Process jobs
php artisan schedule:work	# Run scheduler locally
php artisan tasks:send-reminders	# Manual due-date reminders
php artisan route:list	# All routes
php artisan tinker	# REPL
npm run dev	# Frontend assets (dev)
npm run build	# Frontend assets (prod)

API Usage

Create a token in Tinker:

```
$user = User::where('email', 'admin@taskforge.test')->first();
$token = $user->createToken('api')->plainTextToken;
```

```
curl -H "Authorization: Bearer YOUR_TOKEN" \
-H "Accept: application/json" \
http://localhost:8000/api/tasks
```

Running Tests

```
php artisan test
# 14 tests covering auth, tasks, and API
```

Deployment

See [deploy/](#) for Nginx, Supervisor, and GitHub Actions configs.
Full deployment guide: [docs/16-deployment-hosting.md](#)

Learning Path

Work through modules in order, implementing each feature as you go:

1. **Modules 00-02** - Setup, PHP, lifecycle
2. **Modules 03-06** - Routes, views, database, auth
3. **Modules 07-11** - API, assets, validation, storage, queues
4. **Modules 12-15** - Events, cache, testing, security
5. **Modules 16-18** - Deploy, monitor, advanced architecture

Start here: [docs/00-getting-started.md](#)